

Step 14: Rainfall vs. River Level Analysis Summary

(Storm event: 11 to 12 November 2025)

Stations used:

- > Rainfall- station 47112 (15-min intervals)
- > River level- Newnham Park, station 47119 (Tory Brook)
- > River level- Carnwood, station 47121 (River Plym)

Did this event align with an official flood alert?

Yes, it did align with official flood alerts. The Environment Agency issued a flood alert on the evening of the 11th of November 2025, naming the River Plym, Tory Brook and Long Brook, warning of potential flooding to low-lying land and roads near Plympton & Marsh Mills from around 9pm that evening, in response to heavy and persistent rainfall. The recorded data shows the heaviest rainfall arrived a few hours later than the alert's initial 9pm window, overnight into the early hours of the 12th of November. Peaking early that morning. This is a reasonable real world pattern. Alerts are issued ahead of the rainfall they're responding to & the most intense rain can arrive later than the alert's original timing estimate.

Source: Plymouth City Council news release, "Weather warnings and flood alert," 11 November 2025.

Rainfall summary:

- Total rainfall over the 2-day window: 29.5 mm

Two distinct pulses:

- Evening of 11 Nov (~16:30–18:45): moderate, ~6 mm
- Overnight into 12 Nov (~02:30–05:15): heavier, ~11 mm, peaking at 2.7 mm/15min at 04:30

River level response:

Station	Baseline (m)	Peak (m)	Peak time	Rise
Newnham Park (Tory Brook)	~0.50	0.616	12 Nov, 05:45	+0.12 m
Carnwood (River Plym)	~0.56	1.103	12 Nov, 07:15	+0.54 m

Lag time (rainfall peak -> level peak):

- Newnham Park (Tory Brook): 1h 15min
- Carnwood (River Plym): 2h 45min

Cross-correlation analysis confirms these lags are a real signal ($r \approx 0.45$ for Newnham Park, $r \approx 0.53$ for Carnwood), not just a coincidence.

Interpretation:

The Tory Brook sub-catchment at Newnham Park responded quickly, peaking roughly 1h 15min after rainfall intensity peaked. While the larger integrated River Plym catchment at Carnwood peaked almost an hour and a half later still (2h 45min). This is consistent with catchment scale governing response time- the smaller & steeper Tory Brook catchment is "flashier," while the larger Plym catchment, draining a wider and more varied area, takes longer to concentrate flow at the gauge. The response aligns with the EA's flood alert issued the evening before, which anticipated impacts from the heavier overnight rainfall rather than the smaller evening pulse.

Methodology notes:

An initial effort at this analysis used data from 4 April 2026, a day with very little rainfall (2.55 mm total). That dataset showed sudden step changes in river level with no apparent rainfall driver, more consistent with sensor artefacts than actual catchment response, and was discarded in favour of this documented storm event. This is a useful illustration of why event selection matters when analysing rainfall response relationships: a little, real signal from actual rainfall is significantly analytically useful than an unclear one from a quiet day.

Files in this set:

- 'tory_brook_rainfall_vs_level_chart.png'- dual-axis chart (rainfall bars, inverted, vs. river level lines for both stations)
- 'tory_brook_storm_merged_data.csv'- cleaned, timestamp-aligned dataset (15-min intervals, 11–12 Nov 2025)

-> Data source: Environment Agency real-time flood-monitoring API / daily archive dumps (environment.data.gov.uk/flood-monitoring), Open Government Licence v3.0.